Gen AI

AI tools can be used in an assistive role

COMP1850 Programming

School of Computer Science, University of Leeds

Worksheet - Week 2.1

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio.

- Portfolio activities are indicated in the title of the task.
- Activities marked by (*) are advanced activities and may take more time to complete.

Expectations:

1. Timeliness: You should complete all the tasks in the order provided and submit your portfolio task to Gradescope before the deadline.
2. Presentation: You should present all your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
3. Integrity: You are responsible that the evidence you submit as part of your portfolio evidence is **entirely your own work**. You can find out more about Academic integrity on the Skills@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.
Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via office hours.
Expected complete date: Friday 6th February

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment.

Within this assessment you **may** use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries
- test your knowledge against a piece of content
- help identify and correct spelling mistakes and grammatical errors in your work
- provide feedback and advice on your overall coding style

You must **not** use Gen AI to:

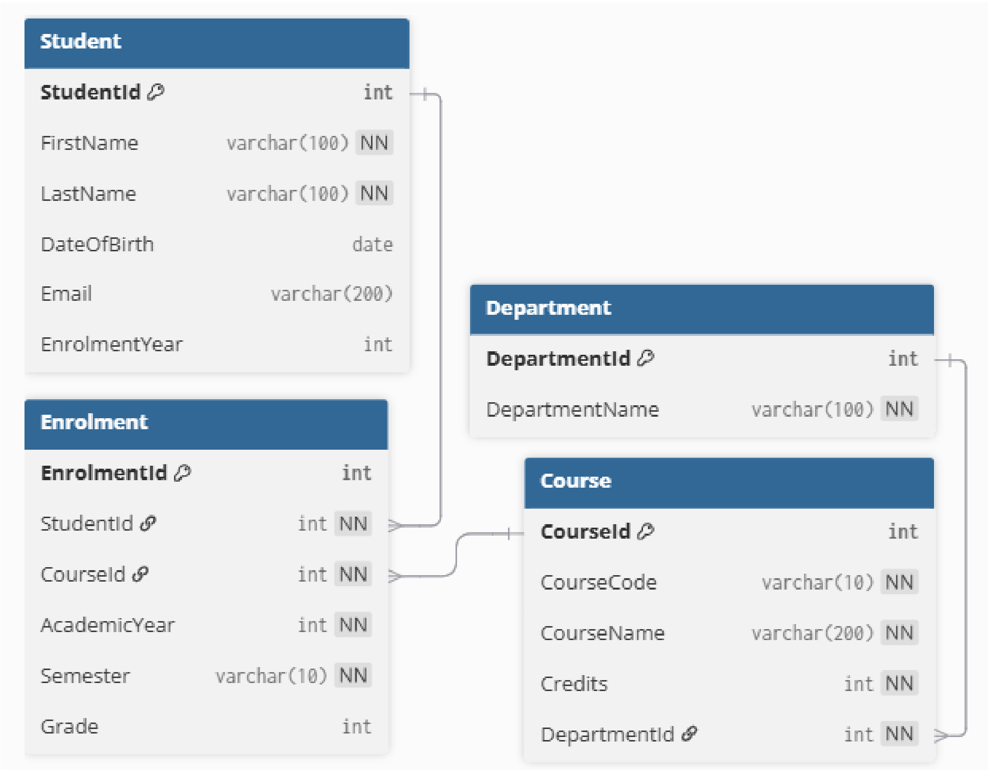
- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning Outcomes:

- Compose SQL queries to retrieve data from a database
- Identify the common normal forms.
- Articulate the components of a relational data model

Task 1

Look at the database diagram for student.db:



The following are exam style questions – try and provide a suitable exam-style answer for each. The number of marks which would be given for each question is in [square brackets] after the question.

1. Identify two entities shown in the database diagram and briefly describe what real-world objects they represent. [4]
2. Define the term primary key with reference to one primary key from the database. [2]
3. Explain what is meant by a foreign key. Using the database diagram, identify one foreign key and describe the relationship it enforces. [3]
4. Describe how the database models the relationship between students and courses. Explain why this relationship cannot be represented using a single table. [3]
5. Explain the purpose of the Enrolment table. Identify two attributes stored in this table and justify why they belong there rather than in another table. [4]
6. Identify the highest normal form that this database satisfies and justify your answer with reference to the structure of the tables and their attributes. [4]

If you would like to answer digitally, these questions are provided in digital form in the worksheet folder.

Task 2 – portfolio

On Github in the task2 folder, tasks.md contains 5 questions.

You must write an SQL query for each question, and upload these to Gradescope where they will be run by the autograder.

Each question specifies the **exact** columns which should be selected – including names where these need to be aliased using AS. Ensure that you are selecting the correct columns.

Your files must be uploaded with the exact names (question1.sql, …) which they have on Github – there are instructions in the tasks.md file for how to download them in a zip which will preserve names and can be drag-and-dropped into Gradescope.

Task 3 (*) – Optional Challenge

On Minerva, you will find Q3 – F1 Quiz.

This is a quiz with two halves – the first is some general questions about SQL, and the second needs you to extract some data from the formula1.sqlite database which is provided in task3/.

This is a large and complex database – a diagram is provided on Github.

You will need to think carefully about what tables need to be joined to get the right information – some good practice if you enjoyed working with the larger databases this week.

You can resubmit as many times as you want – the answers will be displayed after the due date.